Islamic University of Technology

Lab 8: Arrays

CSE 4108 : Structured Programming Lab

Section 1A

MIA, NFS, IAC, ASH

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1 Lab Tasks

1.1 Out with the Outliers

Teachers of IUT CSE want to find the outlier marks of their students. They have the marks obtained by the students in ascending order. Write a program that takes the number of students (n) as input and then scans the marks achieved by each student. Then it prints out the mean mark, median mark (Q2), interquartile range (IQR) and the outliers, if any. Marks are between 0 to 100 and may contain a fractional part. Number of students (n) is given in such a way that n+1 is **always a multiple of 4**.

To find the quartiles, the following formula may be used.

If there are n marks, and n is odd:
$Q1 = [(n+1)/4]$ th mark
$Q2 = [(n+1)/2]$ th mark
$Q3 = [3 * (n+1)/4]$ th mark

If there are n marks, and n is odd,
$Q1 = [n/4]$ th mark
$Q2 = [(n+1)/2 + n/2]/2$ th mark
$Q3 = [3 * n/4]$ th mark

$IQR = Q3 - Q1$

Outliers are any mark (m) such that
$m < Q1 - 1.5 * IQR$ or
$m > Q3 + 1.5 * IQR$.
If there are no outliers, print "Outlier(s): None" instead of the outlier marks.

[Sample Program]
Input

```
Enter student count: 7
Enter marks of each student: 30 40 45 55 56 60 97
```

Output

```
Mean mark: 54.71
Median mark: 55.00
Interquartile range: 20.00
Outlier(s): 97
```

1.2 Selection Sort

Sorting is a very common sub-task in almost any programming applications. One simple way to sort a series of data in an array is to use the Selection Sort algorithm. As the name suggest, the algorithm Selects the smallest element from an array and changes its position. A pseudo-code for selection sort for sorting is given here.

You may search the internet to find more relevant materials and description on selection sort. There are many ways to implement selection sort. However, do not copy from the internet without properly understanding what is happening.

Now, write a program that would take an array length and then the values of the array as input from the user and then print out the sorted array in ascending order using selection Sort. You may use VLAs or consider that the length of the array will not exceed 200.

[Sample Program]

Input 1

```
Enter array length: 5
Enter values for the array: 1 3 2 -1 2
```

Output 1

```
Sorted array: {-1, 1, 2, 2, 3}
```

Input 2

```
Enter array length: 8
Enter values for the array: 9 8 7 6 5 4 3 2
```

Output 2

```
Sorted array: {2, 3, 4, 5, 6, 7, 8, 9}
```

[Hint] *Selection sort sorts the array in place, meaning it modifies the original array.*

1.3 Light Spiral

After staying home for two years, Light Origami lost his abilities to function as a normal human. He now reads all matrices in a clockwise spiral way. His friends are trying to understand how Light looks at the world.

Help his friends by writing a program in C that would scan an integer `n` from the user and then scan an `n`×`n` matrix. The output will be the matrix printed in Spiral form.

You may use VLAs or consider that `n` will not be more than 10.

[Sample Program]

Input

```
Enter dimension N: 4
Enter Matrix [4x4]:
 1   2   3   4
12  13  14   5
11  16  15   6
10   9   8   7
```

Output

```
Matrix that Light reads[4x4]:
|       1    2    3    4    |
|       5    6    7    8    |
|       9   10   11   12    |
|      13   14   15   16    |
```

1.4 Longest Substring

Write a program that would take a sentence and a letter as input from the user and then print out the longest substring consisting of that letter in the sentence and the index where it occurs.

Do not use strings. Rather read manually until newline and store the sentence and the word in arrays. You can consider that the sentence will not be larger than 100 characters. The system should be case sensitive.

[Sample Program]
Input 1

```
Enter a sentence: Never gonnnna give you uppp never gonnna let you downn
Enter a letter: n
```

Output 1

```
Substring: nnnn
Starting index: 8
```

Input 2

```
Enter a sentence: Rrrolling with Rick
Enter a letter: r
```

Output 2

```
Substring: rr
Starting index: 1
```

[Hint] *One loop for the sentence, a nested one for the letter occurrences.*